

Course Introduction

STAT 481 — Multivariate Analysis — Fall 2026

Section L01 (CRN 16064) • Sun/Tue/Thu 12:00–12:50 • BCR — Corridor H210

Sun, Aug 23, 2026 • 50 minutes • *first day of classes* — *Lecture 1 is Tue, Aug 25*

Today's Overview

- **The question this course exists to answer:** you have 41 cities and seven measurements on each. Which city is unusual, and unusual *how*? That question is not answerable one variable at a time.
- How the semester runs: 26 lectures, of which 8 are **R labs**; ten practice and revision sessions; four quizzes; a midterm; a project, in pairs.
- The assessment plan and every date.
- **What you must install before Tuesday** — R, RStudio, Quarto — and how to open the course project.

Packages today

Nothing to install for today — but everything below must be working on your own laptop by **Tuesday, Aug 25**. Section 5 tells you exactly how.

- **R** (4.5.2 or newer) **[NEW]** — the language everything in this course is computed in.
- **RStudio** **[NEW]** — the window you actually work in.
- **Quarto** **[NEW]** — turns your code and your writing into one document; this is how every lab report and the project are submitted.
- **HSAUR2** **[NEW]** — the first *package* we use, on Tuesday. A package is a bundle of extra functions and data; step 5 of Section 5 installs it for you.

No prior programming experience is assumed — see Section 3. Every lecture opens with a “Packages today” box like this one.

By the end of today you should be able to

- Say what makes a question *multivariate*, and why it needs its own course.
- Describe how a lecture, an R lab, a practice session and a quiz differ, and what you are expected to do in each.
- State how your grade is calculated and name the date of every assessment.
- Install R, RStudio and Quarto, and open `STAT481.Rproj` without help.

1 Welcome, and the one question

Welcome to STAT 481. I am Prof. Dhafer Malouche (Department of Mathematics and Statistics, BCR-D 203, dhafer.malouche@qu.edu.qa, office hours Tuesdays 10:55–11:55 in BCR-D 203 or by appointment). This is a required course for the B.Sc. in Statistics, it carries 3 credits, and its prerequisites are STAT 322 and MATH 231 — so you have met a hypothesis test about one mean, and you have met a matrix. If either feels distant, that is normal and not a reason to leave: we rebuild both in Weeks 4–5. This course is about what happens when you have *several* measurements on each subject at the same time, and you cannot look at them one at a time without losing the point.

Here is the question we are going to open with on Tuesday, and it is worth stating today because the whole semester is an answer to it.

The question

Forty-one American cities. Seven numbers for each one: sulphur dioxide in the air, average temperature, number of manufacturing enterprises, population, wind speed, annual rainfall, and the number of days a year it rains.

Which of these cities is unusual — and unusual how?

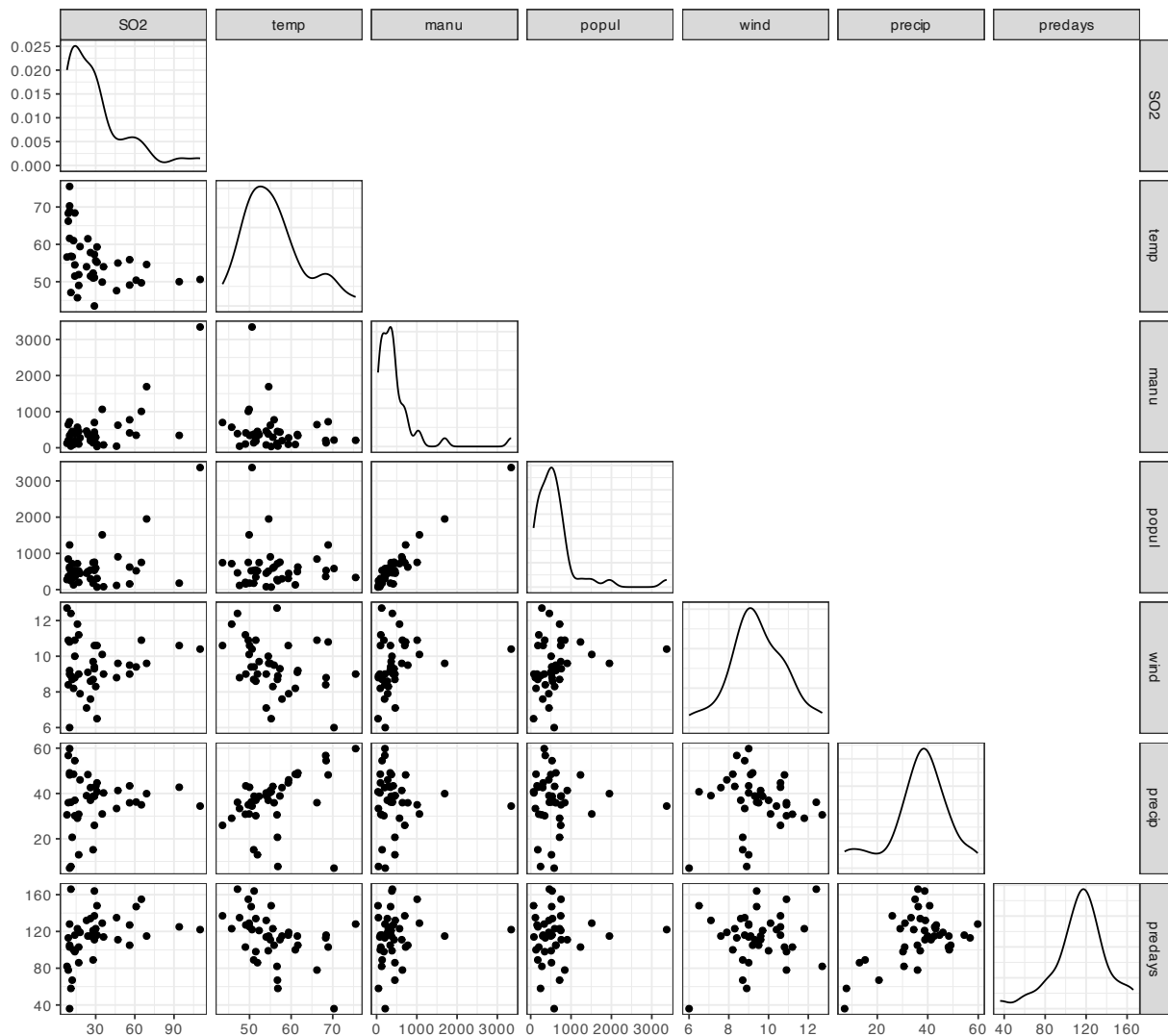
Notice what happens when you try to answer it one variable at a time. “Which city has the most sulphur dioxide?” has an answer, and finding it takes one line of code and no course. But *unusual* is different. A city can be perfectly ordinary on every single one of the seven measurements and still be a very strange *combination* of them — a cold, dry, enormous city, or a hot, wet, tiny one. The moment you ask about the combination, you need a way to talk about seven numbers as one object. That object is what this course is about, and it takes us until **Thursday, Sep 3** to build the tool that answers the question honestly.

The catalog says it like this: the course covers the organisation of multivariate data, multivariate distributions, Mahalanobis distance, Hotelling’s T-squared statistic, multivariate analysis of variance and regression, data-reduction techniques, discriminant and classification analysis, and canonical correlation analysis — with statistical software used throughout on real data. Every one of those words will mean something to you by December.

2 The course in one picture

Below is the entire data set: forty-one rows, seven columns. You are not expected to read it yet — only to notice that it is dense, that some panels clearly have a pattern and others clearly do not, and that no one panel by itself answers the question we just asked.

How to read it. Each small panel plots one pair of variables; the diagonal curves show one variable. The seven short labels are the seven measurements from the box above: **SO2** sulphur dioxide, **temp** temperature, **manu** manufacturing enterprises, **popul** population, **wind** wind, **precip** rainfall, **predays** rainy days. The upper half of the grid is deliberately empty: the usual way to fill it is one number per pair, measuring how tightly the two move together — and we build that number on **Thu, Aug 27**.



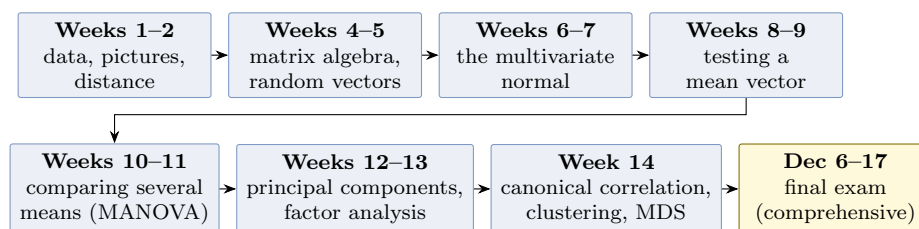
The 41-city air-pollution data, all 21 pairs at once. We meet it properly on Tuesday, and we will still be using it in November.

The promise

By December you will be able to reduce these seven variables to two — and explain what those two mean. That is: replace the seven columns by two new ones built from them that keep most of what made the cities differ — so the grid above collapses into one scatterplot.

That is called principal component analysis, it arrives on **Sun, Nov 15**, and everything between now and then is either the machinery it needs or a method that shares its ideas.

Where the semester goes



Two things are worth saying about that diagram. First, it is genuinely a *sequence*: Weeks 4–5 exist because Weeks 12–13 need them, and if the matrix algebra feels unmotivated at the time, that is the debt being paid forward. Second, every one of those boxes is done on real data, in R, in this room. (MDS is multidimensional scaling: drawing a map from nothing but a table of distances.)

3 How the semester runs

We meet three times a week, Sunday, Tuesday and Thursday, 12:00–12:50, for fourteen teaching weeks — 42 sessions in all. Today is one of them; the other 41 are of four kinds, and it matters that you can tell them apart.

Kind of session	How many	What happens, and what you do
Lecture	18	New material. I build one idea, we do a worked example together, and there is at least one live R chunk and one figure — <i>in every lecture, including the theory ones</i> . You take notes on the printed handout.
R lab	8	Same room, same 50 minutes, but you type. The format is: I demonstrate, then “now you” — three numbered checkpoints you complete during the session, plus a short take-home mini-lab. Bring your laptop.
Practice session	10	We solve a practice sheet together, correct a sample quiz, or revise for the midterm or final, in the exact format of the real thing. This is where you find out what the quiz will look like before you sit it.
Assessment	5	Four quizzes and the midterm, in class, on paper, closed book.

The one thing that makes this course different

Every method in this course is computed, on real data, in R — by you. Derivations are motivated and stated, and then checked numerically rather than proved line by line. That is a deliberate choice, and it has a consequence you should know about now: **roughly two thirds of every quiz asks you to read, write, predict, fix or interpret R code.** Not to run it — all output is printed on the paper — but to understand it. If you skip the labs, the quizzes will hurt.

None of that assumes you have programmed before. The chunks are a few lines long, every function is taught in the lecture that first uses it, and the labs exist so that the first R you ever read is not on a quiz paper.

Every quiz is preceded by one or two practice sessions in which we solve a practice sheet and correct a *sample quiz* of exactly the length, format and difficulty of the real one; the midterm gets two revision sessions in the same spirit. All of it is posted on Blackboard with full solutions.

4 Assessment

Assessment	Weight	When, and what it covers
In-class quizzes ($\times 4$)	15% <i>all four</i>	Thu Sep 10 (Lectures 1–5) • Tue Sep 29 • Thu Nov 12 • Thu Nov 26 . Paper, in class, closed book. There is no make-up for a quiz.
R labs ($\times 5$) and the project	15% <i>together</i>	Lab 1 out Aug 30, due Sep 6 • Lab 2 Sep 17 $\rightarrow$ Sep 24 • Lab 3 Oct 6 $\rightarrow$ Oct 13 • Lab 4 Nov 8 $\rightarrow$ Nov 15 • Lab 5 Nov 17 $\rightarrow$ Nov 24. Project topics Nov 1, report due Dec 1, presentations Dec 3.
Flipped classroom & in-class activities	5%	Ongoing. Selected topics you prepare before class; graded problems during lectures and practice sessions, some submitted via Blackboard.
Midterm exam	30%	Thu, Oct 22, 2026 , in class, covering Weeks 1–8. Revision sessions Oct 18 and Oct 20.
Final exam	35%	Dec 6–17, 2026 , comprehensive. Exact date, time and venue are set by the Registrar, not by me.

Grades are awarded on the University scale: A 90–100, B+ 85–90, B 80–85, C+ 75–80, C 70–75, D+ 65–70, D 60–65, F below 60. Grades are not negotiable.

The four dates to write down now

Quiz 1 Thu, Sep 10	Quiz 2 Tue, Sep 29	Midterm Thu, Oct 22	Quizzes 3, 4 Nov 12, Nov 26
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△! Note the day of the week

Quiz 2 is on a Tuesday, September 29 — not a Thursday like the other three. The Thursday of that week (Oct 1) is needed to start the multivariate normal distribution. Every announcement I make will say “Tue, Sep 29”; if you remember “quizzes are on Thursdays”, you will get this one wrong.

5 Before Tuesday: install R, RStudio and Quarto

This is the only homework from today, and it is not optional: **Lecture 1 on Tuesday contains live R**, and **Lecture 3 on Sunday Aug 30 is a full lab in which you type**. R Lab 1 goes out that day and is due a week later. Give it an unhurried hour tonight, and do the five steps in order.

1. **Install R** (version 4.5.2 or newer) from <https://cran.r-project.org>. Choose your operating system, download, install with all the defaults.
2. **Install RStudio Desktop** (free) from <https://posit.co/download/rstudio-desktop/>. R is the engine; RStudio is the dashboard. You need both, and in that order.
3. **Install Quarto** from <https://quarto.org/docs/get-started/>. Every lab report and the project are submitted as a Quarto PDF.
4. **Download the course project folder** from Blackboard. It arrives as a .zip: *unzip it first* — working inside the zip preview window is the commonest way this step fails. Then double-click **STAT481.Rproj** in the unzipped folder. RStudio opens with that folder as its *working directory* — the one place R looks when you ask it for a file by name.

5. In the RStudio *console* — the panel with a `>` prompt — type this exactly, quotation marks and all, then press Enter:

```
source("R/setup.R")
```

Read it as “run the file `setup.R` in the folder `R` inside the project”; the quotes mark a file name, and the slash works on Windows too. It reports which course packages you have, and for anything missing prints a complete `install.packages(...)` line with the real names filled in. Copy *that* line, not the literal `(...)`; run it, then `source` again until nothing is missing.

△! Two things that cost marks all semester

1. Always open the project, never `setwd()`. (`setwd()` is R’s “switch to this folder” command; you will meet it in code online.) Double-clicking `STAT481.Rproj` sets the working directory correctly for you. A path like `/Users/you/Desktop/stat481` exists on your machine and on no one else’s, so a report that relies on it will not run on mine.

2. Your report must load its own data and attach its own packages. Nothing to do tonight: this starts with Lab 1, and “render” is the Quarto button that makes the PDF you upload. Before submitting anything: restart R (`Session` → `Restart R`), render the document from that fresh session, and check that the PDF you upload is the one that just came out. This is the single most common way to lose the reproducibility marks.

If it does not work

Come to office hours (**Tuesdays 10:55–11:55, room BCR-D 203**) or catch me after class **before** Sunday Aug 30. An installation problem is a five-minute conversation and a lost week if you sit on it. Read the *first* error line, not the last one — it usually names the actual problem.

6 Course regulations

Qatar University — course regulations

Attendance. You are expected to attend all classes. **A student who misses more than 25% of the classes will receive a failing grade regardless of performance.** The attendance record begins on the first day of class, irrespective of the Add/Drop period. It is your responsibility to sign in and to monitor your own absences.

Calculators. A scientific calculator is allowed in quizzes and exams — anything with power, root, logarithm and exponential functions (Casio FX-82 ES Plus or equivalent). **Graphing calculators are not permitted**, and calculators may not be shared during an assessment.

Quizzes and exams are closed book. There is **no make-up for a quiz**. If you miss an assessment without prior permission and without a medical note, that component is a fail.

Late lab reports and project. 10% penalty immediately; a further 10% after 24 hours, and a further 10% for each subsequent 24 hours.

Plagiarism is a serious academic offence: using someone else’s work, words, images or ideas without acknowledgement, or misrepresenting your sources. I may fail the coursework or deduct marks where it is detected.

Qatar University — course regulations (continued)

Use of AI tools. Quizzes, the midterm and the final are closed-book in-class assessments: **any use of AI tools during them is academic misconduct**. For lab reports and the project, AI may be used only as a study aid — to check reasoning or explore R code — and **any such use must be acknowledged in one line at the end of the report**. The submitted work must be yours, in your own words, showing complete derivations, output and justification; answers without visible work receive no credit. **You may be asked to explain any part of your own submitted report**, in class or in office hours, and being unable to do so costs the corresponding marks.

△! Today is day one of the attendance record

The 25% rule counts from **today**, Sun Aug 23 — not from the end of Add/Drop. Three sessions a week for fourteen weeks is 42 sessions, so 25% is about **ten missed sessions**. That sounds like a lot in August. It is not.

7 Materials

Not covered in class — read this section tonight.

Required textbook. Richard A. Johnson and Dean W. Wichern, *Applied Multivariate Statistical Analysis*, 6th edition, Pearson. I will refer to it as “J&W” and give section numbers with each lecture.

The R companion. B. Everitt and T. Hothorn, *An Introduction to Applied Multivariate Analysis with R*, Springer 2011 — this is where the `USairpollution` data comes from, and it is the closest published book to how this course actually works.

Also useful: Rencher, *Methods of Multivariate Analysis* (2nd ed.); Härdle & Simar (3rd ed.); Manly, *A Primer*; Anderson (3rd ed.).

On Blackboard: the lecture notes L1–L26 as you are reading this one, every practice sheet and sample quiz with solutions, all R and Quarto scripts, and the course project folder.

Self-check (1 minute)

Three questions, and you should be able to answer all three before you leave.

- (a) Which two assessments are together worth 65% of your grade, and when are they?
- (b) On what day of the week is Quiz 2?
- (c) What are the three pieces of software you must have working by Tuesday?

*(Answers: (a) the midterm, 30%, Thu Oct 22, and the final, 35%, in Dec 6–17 — which is why the four quizzes, 15% between them, are rehearsals rather than the grade. (b) **Tuesday**, Sep 29. (c) R, RStudio and Quarto.)*

Wrap-up

Summary — what you must do before Tuesday

1. **Install R, RStudio and Quarto** on the laptop you will bring to class — tonight, not the night before the lab.
2. **Download and unzip the course project**; check that `STAT481.Rproj` opens RStudio cleanly and that `source("R/setup.R")` runs.
3. **Bring your laptop on Sunday, Aug 30** — Lecture 3 is a lab; you will type for 50 minutes.
4. If anything fails, **tell me before Aug 30**, not after.

Next session: Lecture 1, Tue Aug 25 — “Organising Multivariate Data”. We meet the 41 cities, we build the data array, and we run the first R of the course.

Announcements:

- **End of Add/Drop: Thursday, Aug 27, 2026.** If you are deciding whether to take this course, decide by then — and come and talk to me first if the R worries you. Later safety net: the **course-withdraw deadline is Thu, Oct 15** — after two quizzes and two R labs.
- **R Lab 1 is out Sun, Aug 30** and is due **Sun, Sep 6**.
- **Quiz 1 is Thu, Sep 10**, covering Lectures 1–5.
- Attendance is recorded from today.

Before Tuesday (not graded — but the answers are the first thing we use)

Home Exercise 1. Find a data set you already care about — another course, a project, a newspaper, anything. Write down two numbers: how many *units* it has (rows: people, cities, days) and how many *variables* (columns). Then write one question about it that you could *not* answer from a single column.

Home Exercise 2. Install R and RStudio, open the console, and type `2 + 2` followed by Enter. Then type `R.version.string` — *with no brackets after it*, because it is a stored value, not a command to run — and write down exactly what it prints. Bring that line to class on Tuesday.

Answers: 1) No single right answer — if it can be answered from one column it is not yet multivariate; make it harder. 2) [1] 4 — the [1] is not the answer, it is R labelling the first item it prints — then something like "R version 4.5.2 (2025-10-31)"; that date should not be older than 2025.